

Organic Chemistry Practical 3: Synoptic Organic Analysis & Deductive Identification

Multi-Step Structural Elucidation of Unknowns from Bench Microscale Tests and Titrimetric Data

CANDIDATE NAME	URWAH	CENTRE NUMBER	PK082
COMPONENT	Paper 3 – Advanced Practical Skills (Organic Chemistry Suite)	CANDIDATE NUMBER	0142
TIME ALLOWED	2 Hours	TOTAL MARKS	40 Marks

INSTRUCTIONS TO CANDIDATES:

- Answer all questions in the spaces provided on the question paper.
- Give working for all calculations and record all experimental readings to the required precision:
 - Burette readings to the nearest 0.05 cm³; balance readings to 0.01 g / 0.001 g.
 - Thermometer readings to the nearest 0.5 °C.
- You may use an HB pencil for any rough working, graphs, or apparatus diagrams.
- Non-exact numerical answers should be given to three significant figures unless stated otherwise.
- You will need a scientific calculator and the Qualitative Analysis Notes provided on pages at the back.

LABORATORY INVESTIGATION OVERVIEW:

This laboratory pack contains authentic Cambridge Advanced Practical examinations. Each practical scenario provides complete procedures, chemical specifications (FA codes), hazard warnings, structured data collection tables, graphical plotting grids, and comprehensive multi-step mathematical and observational deductions.

Question 1: Structural Elucidation of Four Isomeric C₄H₁₀O Alcohols: FA 1, FA 2, FA 3, and FA 4

Cambridge Syllabus Reference: 9701/31/M/J/23/Q3 · [20 Marks]

Synoptic Deductive Problem:

Four unlabelled colourless liquids, FA 1, FA 2, FA 3, and FA 4, are the four structural isomers of formula C₄H₁₀O.

The possible structural isomers are:

- Butan-1-ol: CH₃CH₂CH₂CH₂OH (primary)
- Butan-2-ol: CH₃CH₂CH(OH)CH₃ (secondary)
- 2-Methylpropan-1-ol: (CH₃)₂CHCH₂OH (primary branched)
- 2-Methylpropan-2-ol: (CH₃)₃COH (tertiary)

Diagnostic Bench Tests Carried Out:

- Test 1: Warm with acidified potassium dichromate(VI) at 60°C.
- Test 2: Warm with alkaline aqueous iodine (I₂ in aqueous NaOH, tri-iodomethane reaction).
- Test 3: Distill the oxidation product of primary/secondary isomers and test the distillate with Tollens' reagent.
- Test 4: Add concentrated HCl and anhydrous zinc chloride (Lucas test) at room temperature.

(a) The results of the bench tests are summarized below. Complete the deduction column in the table. [6]

Test	Observations	Deductions
FA 1: • Acidified K ₂ Cr ₂ O ₇ turns green. • Alkaline I ₂ : no precipitate. • Distilled oxidation product gives silver mirror with Tollens'.	Color change: orange to green. No yellow precipitate. Shiny silver mirror formed.	
FA 2: • Acidified K ₂ Cr ₂ O ₇ turns green. • Alkaline I ₂ : bright yellow crystalline precipitate with antiseptic medicinal smell. • Oxidation product gives no silver mirror.	Color change: orange to green. Yellow crystalline precipitate formed. Solution remains clear with Tollens'.	
FA 3: • Acidified K ₂ Cr ₂ O ₇ turns green. • Alkaline I ₂ : no precipitate. • Distilled oxidation product gives silver mirror with Tollens'. • Lucas test: remains clear after 30 min.	Color change: orange to green. No yellow precipitate. Shiny silver mirror formed. No cloudiness at room temp.	
FA 4: • Acidified K ₂ Cr ₂ O ₇ remains orange. • Alkaline I ₂ : no precipitate. • Lucas test: immediately turns cloudy forming two layers within 1 minute.	No color change (solution remains orange). No precipitate. Instant cloudiness/phase separation.	

(b) Draw the displayed formula and state the systematic IUPAC name for each of the four isomeric compounds FA 1, FA 2, FA 3, and FA 4. [6]

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(c) Explain why FA 4 does not react with acidified potassium dichromate(VI). [2]

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(d) Write the chemical formula of the yellow crystalline precipitate formed in the alkaline iodine test with FA 2 and write an ionic or balanced equation for its formation. [3]

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(e) One of the four isomers exhibits optical isomerism (enantiomerism). Identify this isomer, state the feature responsible, and draw 3D wedge-and-dash representations of the two enantiomers. [3]

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Question 2: Deductive Identification of an Unknown Bifunctional Organic Acid (FA 5)**Cambridge Syllabus Reference: 9701/32/M/J/22/Q3 · [20 Marks]**

Synoptic Analytical Investigation:

An unknown liquid compound, FA 5, contains carbon, hydrogen, and oxygen only.

Elemental analysis reveals that FA 5 contains 55.8% C, 7.0% H, and 37.2% O by mass.

Experimental Observations on FA 5:

- Observation 1: When 1 cm³ of bromine water, Br₂(aq), is added to 5 drops of FA 5, the orange color decolourizes immediately without requiring any catalyst.
- Observation 2: When solid sodium hydrogencarbonate, NaHCO₃, is added to FA 5, rapid effervescence is observed and the evolved gas turns limewater milky.
- Observation 3: A 1.720 g sample of FA 5 was dissolved in distilled water and made up to 250.0 cm³ in a volumetric flask. Titration of 25.0 cm³ portions of this solution required 20.00 cm³ of 0.100 mol dm⁻³ NaOH for complete neutralization using phenolphthalein indicator.
- Observation 4: FA 5 displays stereoisomerism and exists as a pair of geometric isomers.

(a) Calculate the empirical formula of FA 5 from its elemental percentage composition. [Ar: C = 12.0, H = 1.0, O = 16.0]
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(b) Using the titration data, calculate the relative molecular mass, Mr, of FA 5 (assuming it is a monoprotic acid), and determine its molecular formula. [4]

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(c) State the two functional groups present in FA 5 based on Observations 1 and 2. [2]

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(d) Deduce the structural formula of FA 5 and write its systematic IUPAC name. [2]

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(e) Explain why FA 5 exhibits geometric (cis-trans / E-Z) stereoisomerism. Draw and clearly label the E and Z isomers of FA 5. [5]

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(f) Write the structural formula of the organic product formed when FA 5 reacts with bromine water, Br₂(aq). State the type and mechanism of this reaction. [4]

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Examiner Mark Scheme & Indicative Observation Guide

Confidential Marking Points & Error Tolerances for Organic Chemistry Practical 3: Synoptic Organic Analysis & Deductive Identification

Question 1: Structural Elucidation of Four Isomeric C₄H₁₀O Alcohols: FA 1, FA 2, FA 3, and FA 4 [20 Marks]

Part	Marking Points & Expected Observations	Marks
(a)	FA 1 deduction: Primary alcohol, straight-chain / unbranched (butan-1-ol) [2]. FA 2 deduction: Secondary alcohol containing CH ₃ -CH(OH)- group (butan-2-ol) [2]. FA 3 deduction: Primary alcohol with branch (2-methylpropan-1-ol) [1]. FA 4 deduction: Tertiary alcohol resistant to oxidation (2-methylpropan-2-ol) [1].	[6]
(b)	FA 1: butan-1-ol, displayed formula showing all C-H, C-C, C-O, O-H bonds clearly [1.5]. FA 2: butan-2-ol, displayed formula showing -OH on C2 [1.5]. FA 3: 2-methylpropan-1-ol, displayed formula showing methyl branch on C2 and -OH on C1 [1.5]. FA 4: 2-methylpropan-2-ol, displayed formula showing central C bonded to 3 methyl groups and -OH [1.5].	[6]
(c)	FA 4 is a tertiary alcohol [1]; the carbon atom bonded to the -OH group has no hydrogen atom attached to it (no alpha-hydrogen), so C-C bonds would have to be broken for oxidation to take place [1].	[2]
(d)	Formula: CHI ₃ (tri-iodomethane / iodoform) [1]. Equation: CH ₃ CH ₂ CH(OH)CH ₃ + 4I ₂ + 6NaOH → CH ₃ CH ₂ COONa + CHI ₃ + 5NaI + 5H ₂ O [2] (or ionic: CH ₃ CH ₂ CH(OH)CH ₃ + 4I ₂ + 6OH ⁻ → CH ₃ CH ₂ COO ⁻ + CHI ₃ + 5I ⁻ + 5H ₂ O).	[3]
(e)	Isomer: Butan-2-ol (FA 2) [1]. Feature: Possesses a chiral centre (asymmetric carbon atom, C2) bonded to four different groups: -H, -OH, -CH ₃ , and -C ₂ H ₅ [1]. Drawing: Two non-superimposable tetrahedral mirror-image 3D wedge-dash representations correctly showing the four groups around the central chiral carbon [1].	[3]

Question 2: Deductive Identification of an Unknown Bifunctional Organic Acid (FA 5) [20 Marks]

Part	Marking Points & Expected Observations	Marks
(a)	Moles of C = 55.8 / 12.0 = 4.65 mol [1]. Moles of H = 7.0 / 1.0 = 7.00 mol; Moles of O = 37.2 / 16.0 = 2.325 mol [1]. Divide by smallest (2.325): C = 2.0, H = 3.01, O = 1.0. Empirical formula = C ₂ H ₃ O [1].	[3]
(b)	Moles of NaOH in titre = 0.100 x (20.00 / 1000) = 2.00 x 10 ⁻³ mol [1]. Moles of acid in 25.0 cm ³ = 2.00 x 10 ⁻³ mol; in 250.0 cm ³ = 2.00 x 10 ⁻² mol [1]. Mr of FA 5 = mass / moles = 1.720 g / 0.0200 mol = 86.0 g mol ⁻¹ [1]. Empirical formula mass of C ₂ H ₃ O = 2(12.0) + 3(1.0) + 16.0 = 43.0. Multiple = 86.0 / 43.0 = 2. Molecular formula = C ₄ H ₆ O ₂ [1].	[4]
(c)	Observation 1 (bromine water decolourized): Alkene / Carbon-carbon double bond (C=C) [1]; Observation 2 (effervescence with NaHCO ₃): Carboxylic acid group (-COOH) [1].	[2]
(d)	Structural formula: CH ₃ -CH=CH-COOH [1]; Systematic name: But-2-enoic acid (or 2-butenic acid) [1].	[2]
(e)	Condition 1: Restricted rotation about the carbon-carbon double bond (C=C) due to electron density in the pi bond preventing free rotation [1]. Condition 2: Both carbon atoms of the double bond have two different groups attached (one has -H and -CH ₃ ; the other has -H and -COOH) [1]. Z-isomer (cis): High-priority groups (-CH ₃ and -COOH) on the same side of the double bond [1.5]. E-isomer (trans): High-priority groups (-CH ₃ and -COOH) on opposite sides of the double bond [1.5].	[5]
(f)	Product formula: CH ₃ -CH(Br)-CH(Br)-COOH (2,3-dibromobutanoic acid) [2]. Type of reaction: Addition [1]. Mechanism: Electrophilic addition (involving polarization of Br ₂ and formation of a cyclic bromonium or carbocation intermediate) [1].	[4]